



ECEN 314 - Signals and Systems

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Module 9 - Fourier Series

Learning Objectives



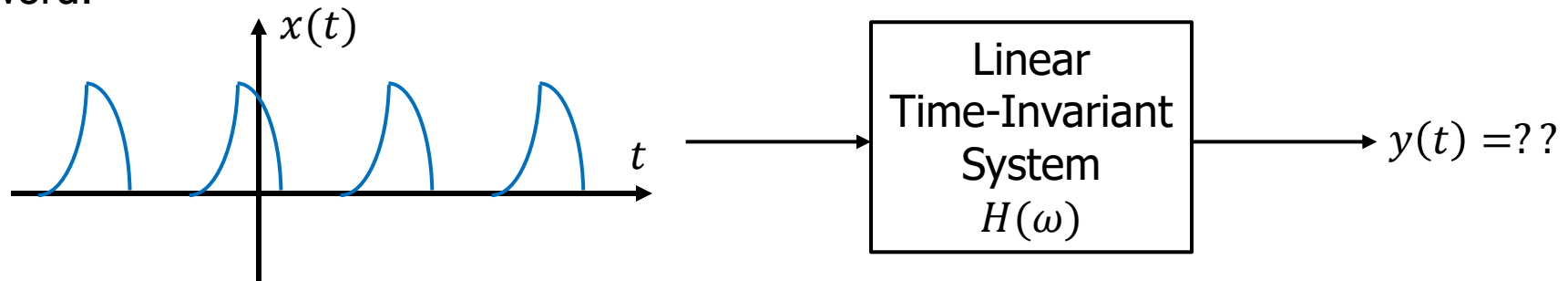
- Given a periodic waveform, be able to determine the period and the fundamental frequency.
- Given a periodic waveform, be able to find the exponential Fourier series coefficients.
- Be able to use Fourier series techniques to find the output of an LTI system due to a periodic input.

LTI Systems with Non-Sinusoidal Inputs



In the last set of slides, we found that it is particularly easy to analyze LTI systems with sinusoidal inputs.

But what do we do if the input is not sinusoidal? Other than invoking the “c” word.



In this set of slides, we will investigate how to analyze LTI systems with other types of (non-sinusoidal) inputs. Our approach will be:

1. Write the non-sinusoidal input as a sum of sinusoids.
2. Use our past results together with the linearity property of LTI systems.

We will focus our attention on periodic signals today and then later generalize to non-periodic signals.

Periodic Signals

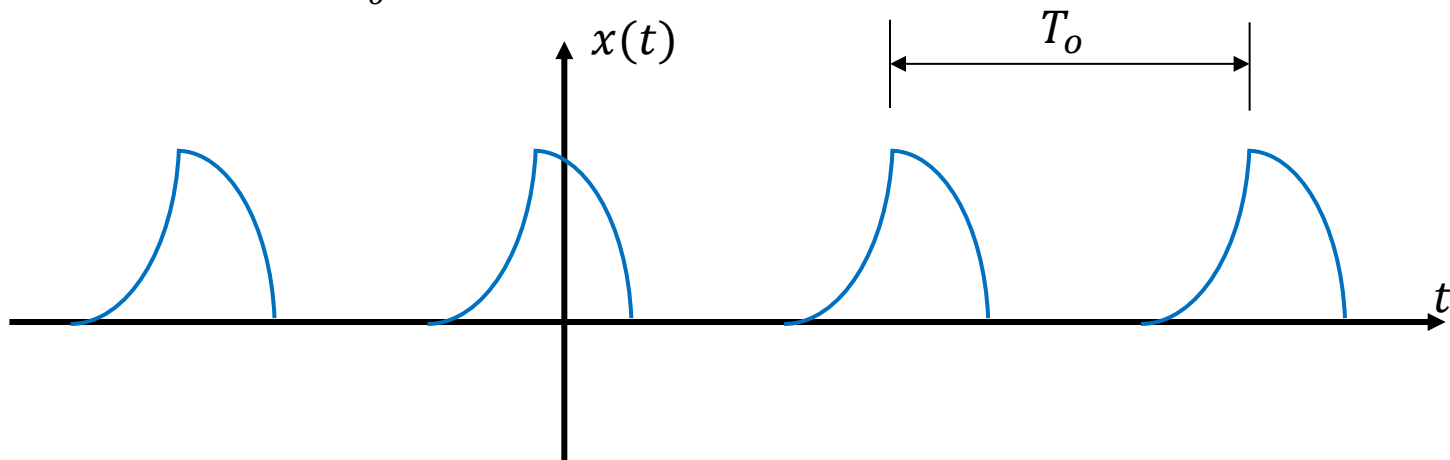


The *period* of a periodic signal, $x(t)$, is the smallest value of T_o such that $x(t + T_o) = x(t)$ for all t .

The *fundamental frequency* (in Hz), f_o , of a periodic signal, $x(t)$, is $f_o = \frac{1}{T_o}$.

The *fundamental frequency* (in radians/second), ω_o , of a periodic signal, $x(t)$, is $\omega_o = 2\pi f_o = \frac{2\pi}{T_o}$.

Multiples of the fundamental frequency are called *harmonic frequencies*, or just *harmonics*, $\omega_k = k\omega_o = \frac{2\pi k}{T_o} \leftarrow k^{\text{th}} \text{ harmonic}$.



Harmonic Sinusoids



- In what follows, we will show how to construct various periodic signals from a series of harmonically related sinusoids.

- It will be convenient (the math works out a little cleaner) to use complex exponentials as our “sinusoids.” In particular, define

$$\varphi_k(t) = e^{j\omega_k t} = e^{jk\omega_o t}, \quad k = \dots, -3, -2, -1, 0, 1, 2, 3, \dots$$

- These harmonically related sinusoids have the property that they are “orthogonal” to one another in the sense that

$$\int_{T_o} \varphi_k(t) \varphi_m^*(t) dt = 0 \quad \text{for all integers } k \neq m.$$

- ✓ The notation $\int_{T_o} \dots dt$ means that the integral is taken over an interval lasting one period. Since the integrand is periodic (with the same period), the starting and ending point of the integral is irrelevant.

- ✓ Note that when $k = m$,

$$\int_{T_o} \varphi_k(t) \varphi_k^*(t) dt = \int_{T_o} |\varphi_k(t)|^2 dt = \int_{T_o} 1 dt = T_o.$$

Fourier Series



- It turns out that any “well-behaved” periodic signal, $x(t)$, can be decomposed into a series of harmonically related sinusoids

$$x(t) = \sum_{k=-\infty}^{\infty} X_k \varphi_k(t) = \sum_{k=-\infty}^{\infty} X_k e^{jk\omega_o t}.$$

- This is known as a *Fourier series* and the appropriate set of constants, X_k , are known as the *Fourier series coefficients*.

- The Fourier series coefficients are found as follows:

$$\begin{aligned} x(t) \cdot \varphi_k^*(t) &= \sum_{m=-\infty}^{\infty} X_m \varphi_m(t) \cdot \varphi_k^*(t) \\ \int_{T_o} x(t) \cdot \varphi_k^*(t) dt &= \int_{T_o} \left[\sum_{m=-\infty}^{\infty} X_m \varphi_m(t) \cdot \varphi_k^*(t) \right] dt \\ &= \sum_{m=-\infty}^{\infty} X_m \int_{T_o} \varphi_m(t) \cdot \varphi_k^*(t) dt \\ &= T_o X_k \end{aligned}$$

The Fourier Series (with complex exponentials) is given by:

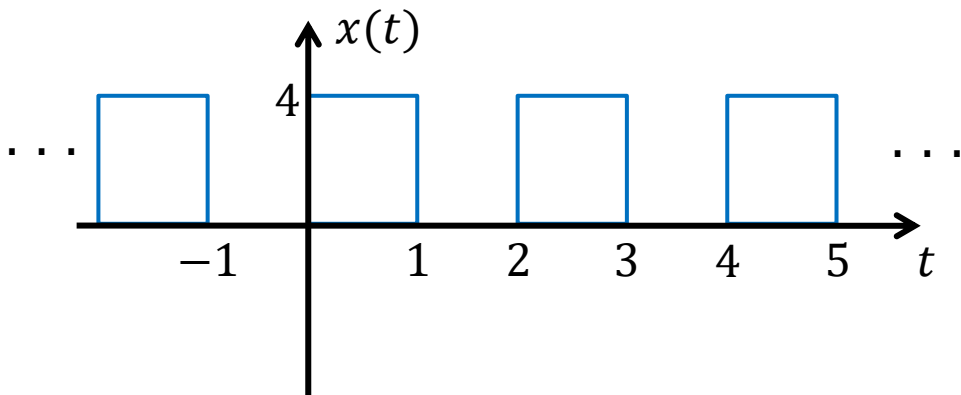
$$x(t) = \sum_{k=-\infty}^{\infty} X_k e^{jk\omega_o t}$$

$$X_k = \frac{1}{T_o} \int_{T_o} x(t) \cdot e^{-jk\omega_o t} dt$$

Fourier Series



Example 9.1: Find the exponential Fourier series coefficients for the waveform shown.



$$\text{Answer: } X_k = \begin{cases} 2, & k = 0, \\ 0, & k \neq 0, \text{ even} \\ -\frac{4j}{k\pi}, & k \text{ odd} \end{cases}$$

Fourier Series



According to the last example, the square wave can be decomposed into a sum of sinusoids as:

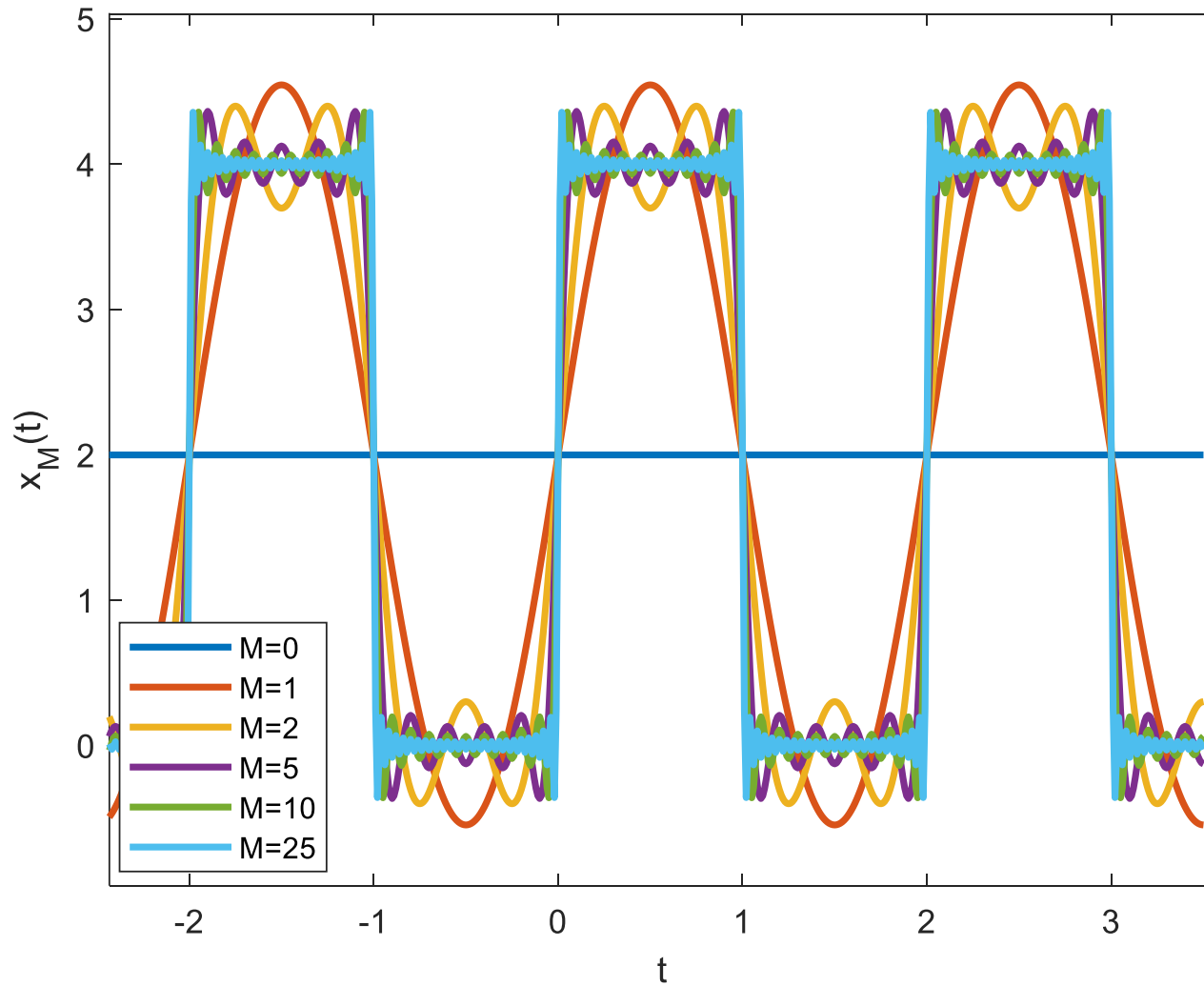
$$\begin{aligned}x(t) &= 2 - \frac{4j}{\pi} \sum_{k \text{ odd}} \frac{1}{k} e^{jk\pi t} \\&= 2 - \frac{4j}{\pi} (e^{j\pi t} - e^{-j\pi t}) - \frac{4j}{3\pi} (e^{j3\pi t} - e^{-j3\pi t}) - \frac{4j}{5\pi} (e^{j5\pi t} - e^{-j5\pi t}) - \dots \\&= 2 + \frac{8}{\pi} \left(\sin(\pi t) + \frac{1}{3} \sin(3\pi t) + \frac{1}{5} \sin(5\pi t) + \dots \right) = 2 + \frac{8}{\pi} \sum_{\substack{k=1 \\ k \text{ odd}}}^{\infty} \frac{1}{k} \sin(k\pi t)\end{aligned}$$

At least for this example, instead of decomposing the square wave into a series of complex exponentials, we could have also written the square wave as the sum of a constant and a series of harmonically related sines.

The picture on the next slide shows how fast (or slow) the series converges. The various curves represent the Fourier series truncated to a finite number of terms:

$$x_M(t) = \sum_{k=-M}^M X_k e^{jk\omega_o t} = 2 + \frac{8}{\pi} \sum_{m=1}^M \frac{1}{2m-1} \sin((2m-1)\pi t)$$

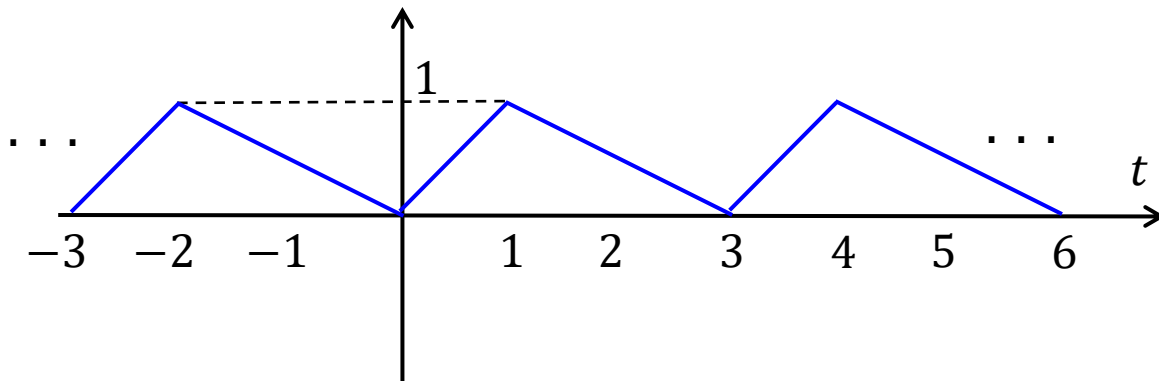
Fourier Series



Fourier Series



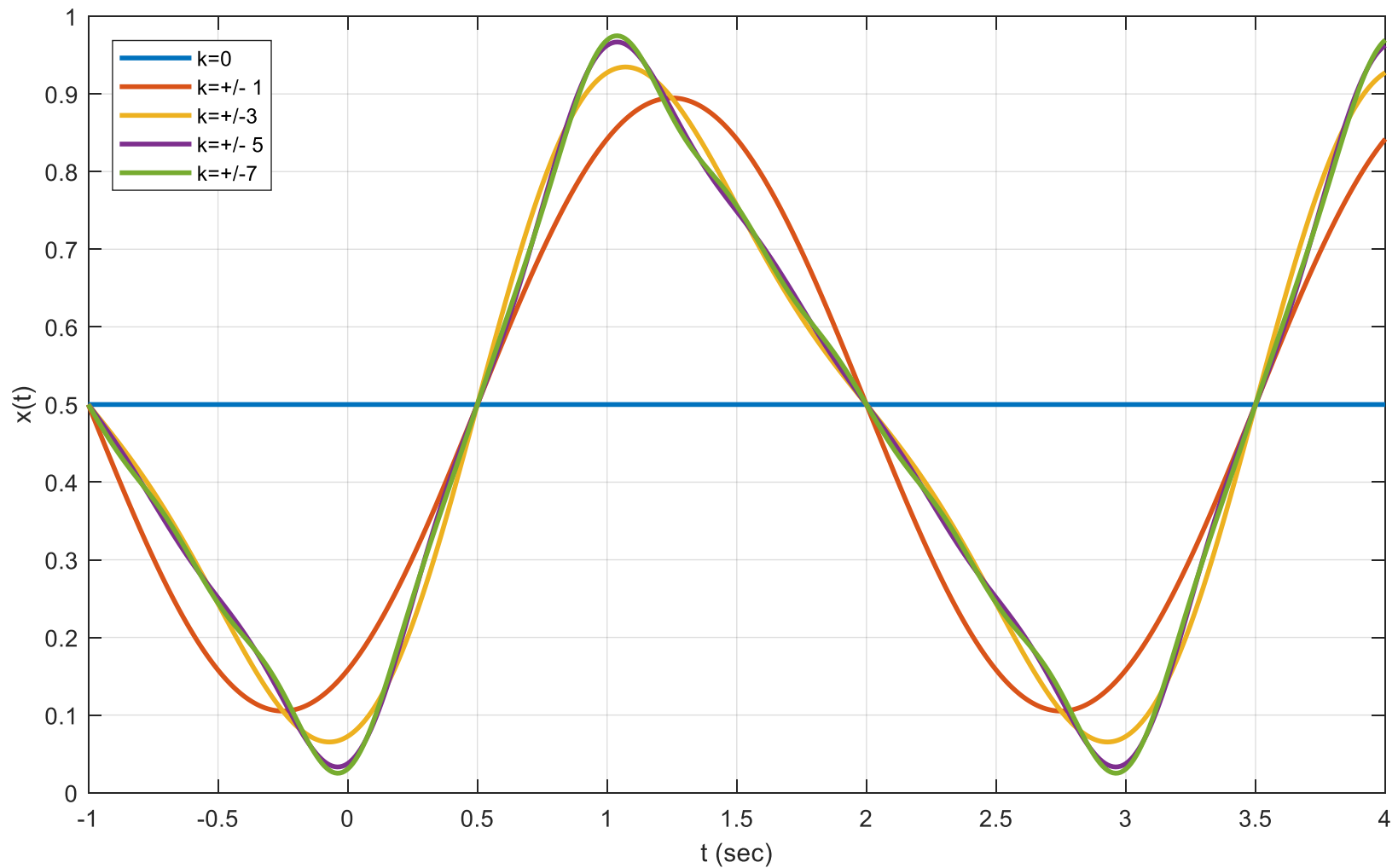
Example 9.2: Find the exponential Fourier series coefficients for the waveform shown.



$$\text{Answer: } X_k = \begin{cases} 1/2, & k = 0, \\ \frac{9}{8\pi^2} \frac{e^{-jk2\pi/3} - 1}{k^2}, & k \neq 0. \end{cases}$$



Fourier Series



LTI System Analysis with Fourier Series

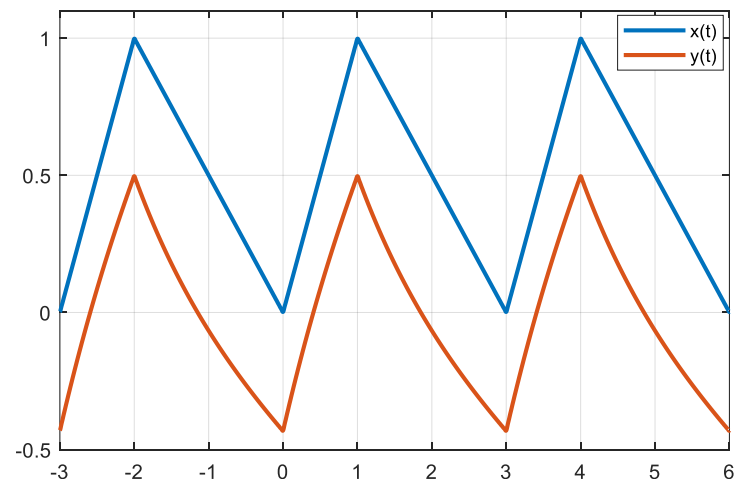
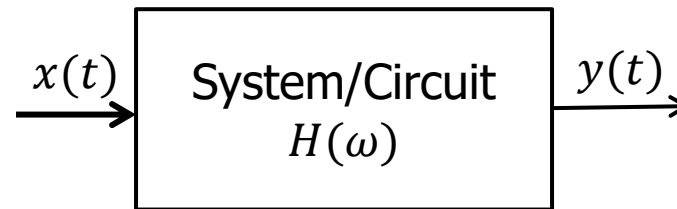


We can use Fourier series to find the response of a circuit/system to inputs which might otherwise be difficult to analyze.

$$x(t) = \sum_{k=-\infty}^{\infty} X_k e^{jk\omega_o t},$$

$$y(t) = \sum_{k=-\infty}^{\infty} Y_k e^{jk\omega_o t},$$

$$Y_k = X_k H(k\omega_o).$$



Example 9.3:

Suppose the circuit is a 1st order (RC) high-pass filter with a cut-off frequency of 1rad/sec,

$$H(\omega) = \frac{j\omega}{1+j\omega}.$$

Let $x(t)$ be the sawtooth wave from Example 9.2 so that

$$X_k = \begin{cases} 1/2, & k = 0, \\ \frac{9}{8\pi^2} \frac{e^{-jk2\pi/3} - 1}{k^2}, & k \neq 0. \end{cases}$$

$$Y_k = \begin{cases} 0, & k = 0, \\ \frac{9}{8\pi^2} \frac{e^{-jk2\pi/3} - 1}{k^2} \frac{jk2\pi/3}{1 + jk2\pi/3}, & k \neq 0. \end{cases}$$

Frequency Content of Periodic Signals



The Fourier series coefficients tell us about the frequency content of periodic signals.

Suppose $x(t) = A\cos(\omega_o t + \theta)$. Decomposing into complex exponentials we get

$$\begin{aligned}x(t) &= \frac{1}{2}e^{j(\omega_o t + \theta)} + \frac{1}{2}e^{-j(\omega_o t + \theta)} \\&= \frac{1}{2}e^{j\theta}e^{j\omega_o t} + \frac{1}{2}e^{-j\theta}e^{-j\omega_o t} \\&\quad X_1e^{j\omega_o t} + X_{-1}e^{-j\omega_o t}\end{aligned}$$

As expected, this sine wave has frequency content at $\omega = \omega_o$ (and at $\omega = -\omega_o$).

Now suppose we squared this signal,

$$\begin{aligned}y(t) = x^2(t) &= \left(\frac{1}{2}e^{j(\omega_o t + \theta)} + \frac{1}{2}e^{-j(\omega_o t + \theta)}\right)^2 \\&= \frac{1}{4}e^{2j(\omega_o t + \theta)} + \frac{1}{2} + \frac{1}{4}e^{-2j(\omega_o t + \theta)} \\&= Y_2e^{j2\omega_o t} + Y_0 + Y_{-2}e^{-j2\omega_o t}\end{aligned}$$

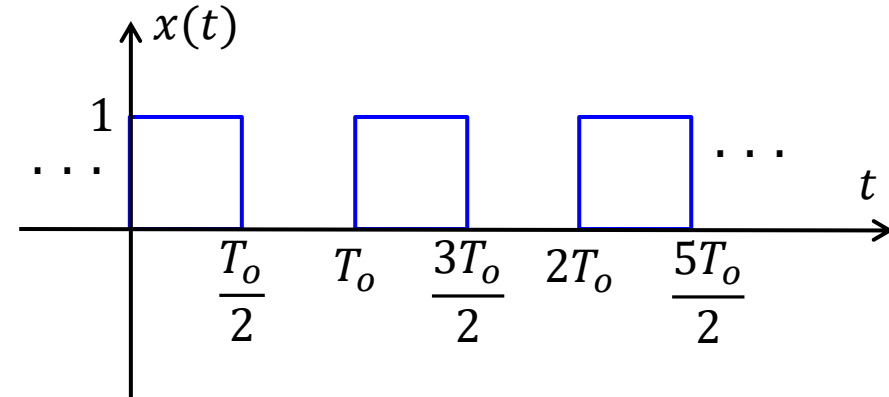
This signal has frequency components at $\omega = \pm 2\omega_o$ and at $\omega = 0$ (DC).

Frequency Content of Periodic Signals

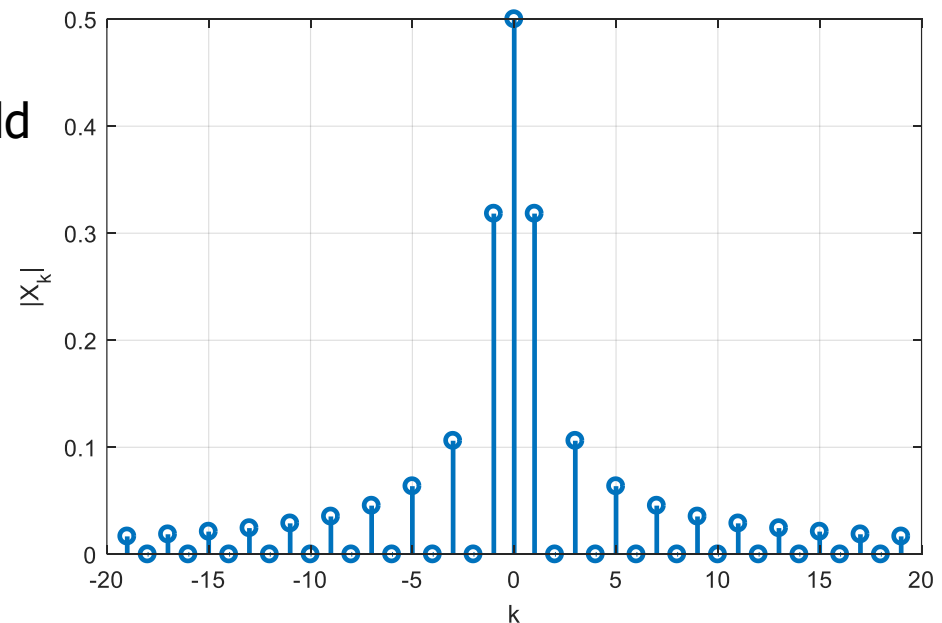


Consider a square wave as shown. The Fourier series coefficients are found to be

$$X_k = \begin{cases} 1/2, & k = 0, \\ 0, & k \neq 0, \text{ even}, \\ \frac{1}{jk\pi}, & k \text{ odd}. \end{cases}$$



- ✓ In addition to the DC component ($\omega = 0$), the square wave has frequency components at all the odd harmonics of the fundamental frequency.
- ✓ All the even harmonics have no frequency content.
- ✓ The strength of the k^{th} frequency component dies off like $1/k$.

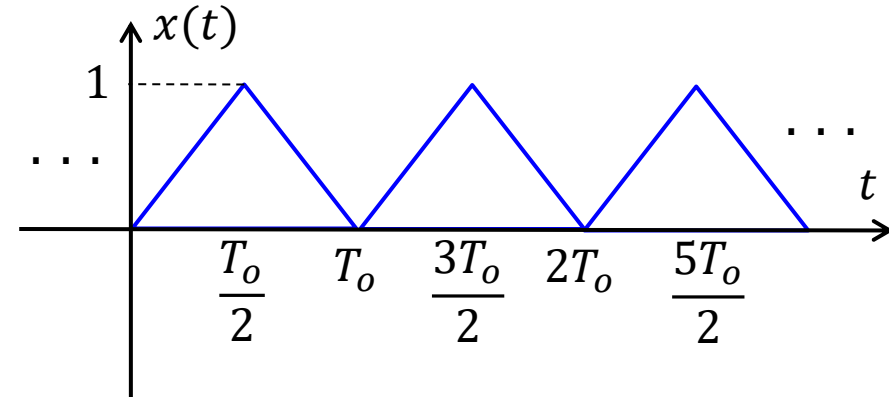


Frequency Content of Periodic Signals

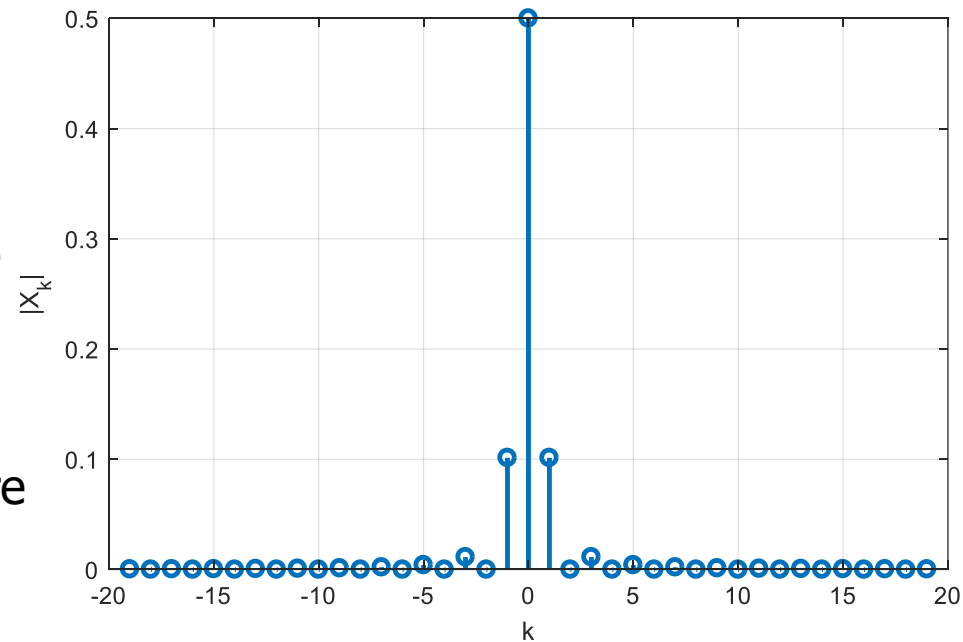


Consider a triangle wave as shown. The Fourier series coefficients are found to be

$$X_k = \begin{cases} 1/2, & k = 0, \\ 0, & k \neq 0, \text{ even}, \\ \frac{-2}{(k\pi)^2}, & k \text{ odd}. \end{cases}$$



- ✓ The strength of the k^{th} frequency component dies off like $1/k^2$.
- ✓ The square wave has more high frequency content than the triangle wave.
- ✓ Signals that are discontinuous (like the square wave) tend to have more high frequency content.



Frequency Content of Periodic Signals



Suppose we pass both the square wave and the triangle wave of the last two examples through an ideal LPF whose cutoff frequency is $6\omega_o$,

$$H(\omega) = \begin{cases} 1, & |\omega| < 6\omega_o, \\ 0, & |\omega| > 6\omega_o. \end{cases}$$

The output due to the square wave will be

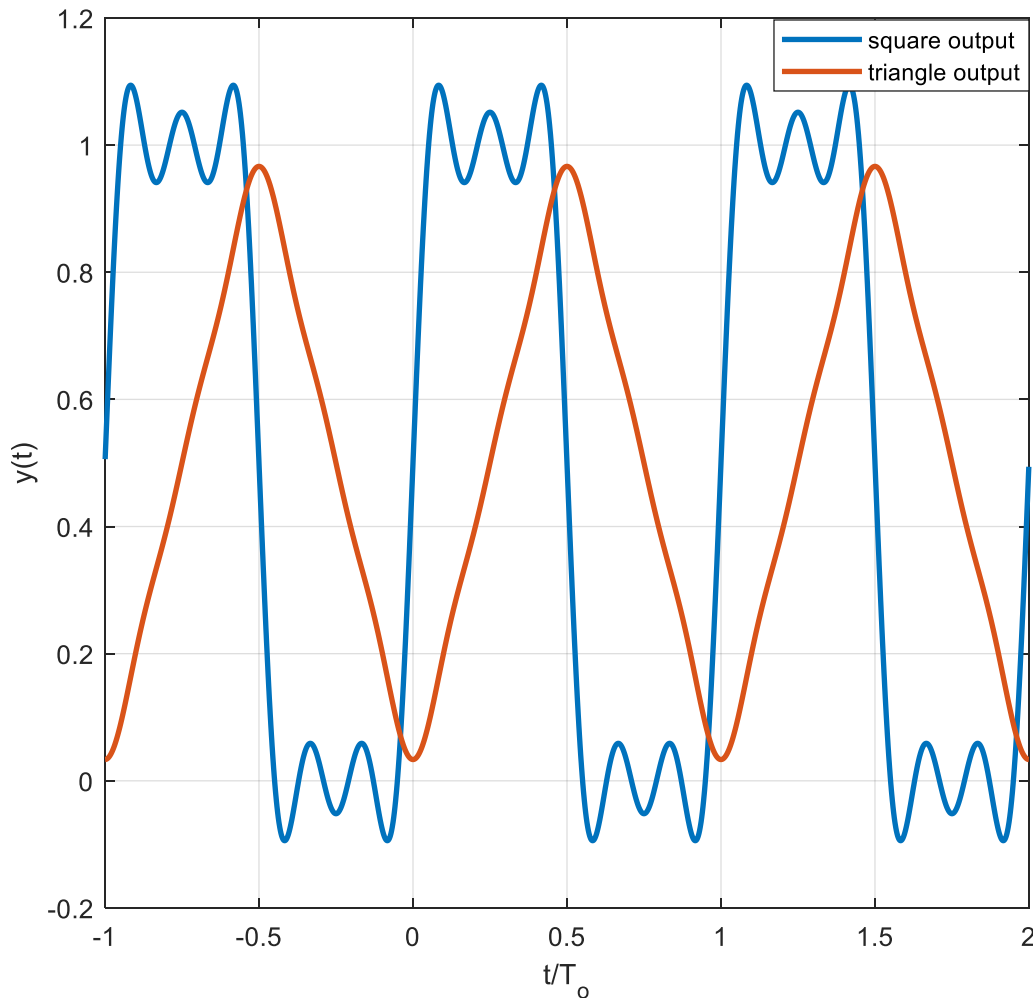
$$\begin{aligned} y(t) &= \frac{1}{2} + \left(\frac{e^{j\omega_o t}}{j\pi} + \frac{e^{-j\omega_o t}}{-j\pi} \right) + \left(\frac{e^{j3\omega_o t}}{j3\pi} + \frac{e^{-j3\omega_o t}}{-j3\pi} \right) + \left(\frac{e^{j5\omega_o t}}{j5\pi} + \frac{e^{-j5\omega_o t}}{-j5\pi} \right) \\ &= \frac{1}{2} + \frac{2}{\pi} \sin(\omega_o t) + \frac{2}{3\pi} \sin(3\omega_o t) + \frac{2}{5\pi} \sin(5\omega_o t) \end{aligned}$$

The output due to the triangle wave will be

$$\begin{aligned} y(t) &= \frac{1}{2} - 2 \left(\frac{e^{j\omega_o t}}{\pi^2} + \frac{e^{-j\omega_o t}}{\pi^2} \right) - 2 \left(\frac{e^{j3\omega_o t}}{(3\pi)^2} + \frac{e^{-j3\omega_o t}}{(3\pi)^2} \right) - 2 \left(\frac{e^{j5\omega_o t}}{(5\pi)^2} + \frac{e^{-j5\omega_o t}}{(5\pi)^2} \right) \\ &= \frac{1}{2} - \frac{4}{\pi^2} \cos(\omega_o t) - \frac{4}{(3\pi)^2} \cos(3\omega_o t) - \frac{4}{(5\pi)^2} \cos(5\omega_o t) \end{aligned}$$

Pictures of these output waveforms are shown on the next slide.

Frequency Content of Periodic Signals



- ✓ These are plots of the output waveforms from the previous slide.
- ✓ The ideal LPF has distorted the square wave much more than the triangle wave because the square wave had much more frequency content outside the pass band of the LPF.